[-1, 0] xor = -1

32 bits.

$\forall$ ✓

1 ✓

excercise

doubts

int {1 → 001, 0 → 000}

-1 → flip → 110 → 110

+1 +001

111

1 → 001

-1 +ve no.

$\forall$

1

xor

111

1 000

MSB

111 → -ve no.

-ve

(111) → 000

+ 1

XOR -(001)

searching

$0 \ldots n-1$ → linear search

i → index

$k \rightarrow$

$0 \ldots n-1$

searching ?

search

find

$K = 6$

how ?

[1 5 2 4 6 3 5]

0 1 2 3 4

4 → ans

Time → WC

complexity

$0 \ldots n-1$

n

O(n)

$k = 5$ $\quad$ $n = 6$

arr = [1 2 4 9 6 5]

ans

$0 \quad i \quad e \quad n-1$

K

```
flag = false;
for(i = 0; i < n; i++){
    if(arr[i] == k){
        print(i);
        flag = true;
        break;
    }
}
if(flag == false)
    print(-1);
```

n $\qquad$ $= O(n)$

$O(n^3)$ $n = 1000$

1 -2 5 -3 2 4 → [1 5 2 3 9]

added

max subarray sum

Kadane's algo

?

gen all subarrays

curr → sum

1st

i = 0 ... n-1

j = i ... n-1

k = i ... j

sum += arr[k]

ans = max(ans, sum);

ans = final

final

(ans)

1 −2 5 −3 2 4

15 → max / 13

+ve + (−ve)

New subarray

sub

$S > 0$ | ϵ

i ... j | j+1

ϵ → −ve element

i ... j

+ve | j+1

$S + \epsilon < 0$

i ... j+1

i ... j+1 | j+2

−ve | x

+ve element

i ... j+2 | j+2 ... n−1

x + (−ve) → x ?

$S = 2 \; -4 \; 6 \; 7 \; 2$

i j j+1 j+2

ϵ

−2

15

| −2 | 6 7 2 |

13

curr → ~~-ve~~ 0

j

j j+1 . . . . . .

-ve

curr = ~~0~~ 1

ans = ~~INT_MIN~~ ~~1~~ ~~5~~ 8

1	-2	5	-3	2	4
0	1	2	3	4	5

$\rightarrow$ once $O(n)$ TC

8

i j j+1 j+2

| s | e | x |

$x > 0$

$s + e < 0$

-ve

curr = 1 - 2

= ~~1~~ ~~0~~ 5 - 3

= 2 + 2

= 4 + 4 = 8

s = 0 + ~~8~~ x

ans